

LECTURE 2.4



Reporting an Estimate

ECONXXXXX Econometrics I

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PLACEHOLDER TERM

Where We Are

1. Foundations

1.1 Data and Variation

1.2 Conditional Means

1.3 Sampling Variation

2. Simple Regression

2.1 Fitting a Line

2.2 Goodness of Fit

2.3 Bias and Precision

2.4 Reporting an Estimate

3. Multiple Regression

3.1 Adding Controls

3.2 Functional Form

3.3 Joint Tests

4. Limits

4.1 Diagnosing Violations

4.2 Limits of Regression

The Last of Four Lectures on One Estimator

Lectures already given are greyed and today's is marked. Lecture 2.1 fitted the line, 2.2 asked how much of earnings it accounts for, and 2.3 asked how far the fitted slope lands from the truth. Today turns it into a reported number.

What Lectures 1.3 and 2.3 Left Open

Lecture 2.3 produced two numbers from our 120 people: a fitted slope of 1.5227 and a standard error of 0.1897. On their own they are not yet a sentence anybody else can check.

Lecture 1.3 turned an estimate and its standard error into a t-statistic, a p-value and a 95 per cent confidence interval. Every one of those was built for a mean.

Today builds all three for a slope. The arithmetic differs in one place only, and it is the degrees of freedom.

Question

Our sample gave 1.5227 euro a year. What else would you want printed beside that number before you would be willing to put it in a report?

Lecture Plan

1 Testing a Coefficient.

Comparing an estimate with its own standard error, and what that settles.

2 Confidence Intervals for an Effect.

A range built from the same two numbers, and whether it covers 1.30.

3 Reading a Regression Table.

What each number in a published table is, and the order to read them in.

SECTION 1

Testing a Coefficient

1. Testing a Coefficient
2. Confidence Intervals for an Effect
3. Reading a Regression Table

Everything Lecture 1.3 Did for a Mean, Now for a Slope

Lecture 1.3 built three objects out of an estimate and its standard error, and none of the three is taught again today:

- **A t-statistic:** the distance from a claimed value, counted in standard errors.
- **A p-value:** what the claim would produce, and never whether the claim is true.
- **A confidence interval:** the estimate and its precision reported as one object.

Wherever Lecture 1.3 wrote $\bar{y}$ and $\text{se}(\bar{y})$, write $\hat{\beta}_1$ and $\text{se}(\hat{\beta}_1)$ instead. The formulas are the same formulas.

Two Things Are Genuinely New

The degrees of freedom are $n - 2$ rather than $n - 1$, because the line used two coefficients rather than one. And the quantity is a slope, so a sentence about it is a sentence about schooling and earnings, in euro per year.

The Standardised Slope Follows a t Distribution

Assumptions one to five of Lecture 2.3 made the slope unbiased and gave its variance. They say nothing about the shape of its distribution, and a test needs the shape:

$$\frac{\hat{\beta}_1 - \beta_1}{\text{se}(\hat{\beta}_1)} \sim t_{n-2} \quad n - 2 = 118 \quad (1)$$

- t_{n-2} : the t distribution on $n - 2$ degrees of freedom
- $n - 2$: the 120 people, less the two coefficients the line already used

One further assumption delivers this: the population error is normally distributed. The slope is then a weighted sum of normal errors and is itself normal, and replacing σ by $\hat{\sigma}$ turns that normal into a t (Wooldridge, 2020).

Chapter 4 Is Stated for Multiple Regression

Wooldridge writes this for k regressors, with $n - k - 1$ degrees of freedom. Our line has one regressor, so $k = 1$ and $n - k - 1$ is 118. He also says plainly that normality is a poor assumption for wages, and that it stops mattering at large sample sizes (Wooldridge, 2020).

The t-Statistic Compares an Estimate With Its Own Standard Error

That result is written in terms of β_1 , which we never see. Putting a claimed value in its place gives a statistic we can compute from our data alone:

$$t = \frac{\hat{\beta}_1 - \alpha}{\text{se}(\hat{\beta}_1)} \quad \text{and, when } \alpha = 0, \quad t = \frac{\hat{\beta}_1}{\text{se}(\hat{\beta}_1)} \quad (2)$$

- α : the value the null hypothesis claims β_1 takes
- t : how many standard errors the estimate sits from that claim

The general form is estimate minus hypothesised value, over standard error (Wooldridge, 2020). An estimate of 1.5227 euro is large or small only against something, and its own standard error is the only comparison the data supplies. The claim is always about β_1 : a null written about 1.5227 is not a hypothesis at all.

Zero Is a Choice, Not a Default

Software prints the test against zero because zero is often the interesting claim, not because it is the right one. If theory says the return should be one euro, subtract one. Wooldridge tests an elasticity

The t-Statistic on Our 120 People

The arithmetic, against a null that schooling carries no return at all:

$$t = \frac{1.5227 - 0}{0.1897} = 8.03 \quad df = 118 \quad c = 1.980 \quad (3)$$

- c : the 97.5th percentile of the t distribution on 118 degrees of freedom

Read out loud: our estimate sits about eight standard errors above zero. The two-sided five per cent rule rejects when $|t|$ is larger than 1.980, and 8.03 is more than four times that (Wooldridge, 2020). Angrist and Pischke's working rule is that a t past about two is already hard to square with the claim (Angrist & Pischke, 2015).

The Wording Is Fail to Reject, Never Accept

We reject a null of zero at the five per cent level. Where a test does not reject, the correct sentence is that the data do not let us reject: many other values survive the same sample, and they cannot all be true (Wooldridge, 2020).

What Rejecting a Null of Zero Does Not Establish

The rejection is a narrow statement: if the true return were zero, a sample like ours would be rare. Four things it is not:

- **Not that the return is 1.5227.** Lecture 2.3 showed the true value is 1.30.
- **Not that the return is large.** 8.03 is a ratio with a precision underneath it.
- **Not that the finding matters.** Whether 1.5227 euro an hour matters is a judgement.
- **Not that schooling causes earnings.** That question is Lecture 4.2's, and no test here touches it.

A p-Value Describes the Claim, Not Its Truth

The p-value is the probability of a t at least this extreme if the null were true (Wooldridge, 2020). It is not the probability that the null is false. Lecture 1.3 settled this for a mean, and nothing about a slope is different.

A Null of 1.30 Rather Than Zero

Zero is not the only value worth testing, and this module knows a more interesting one. Subtract 1.30 from 1.5227 instead, and the numerator becomes the 0.2227 miss from Lecture 2.3, which is a little more than one standard error.

Question

Would a null of 1.30 be rejected at the five per cent level, and what does your answer tell you in advance about the interval we are about to build?

SECTION 2

Confidence Intervals for an Effect

1. Testing a Coefficient
2. Confidence Intervals for an Effect
3. Reading a Regression Table

Reporting a Range Instead of a Point

A t-statistic answers one question about one claimed value and discards everything else the sample had to say. An interval answers it for every claimed value at once, which is why Wooldridge calls intervals *interval estimates*: a range of likely values rather than a point (Wooldridge, 2020).

Lecture 1.3 built one for a mean, as the estimate plus or minus a critical value times its standard error. Today's object is that object, built from a slope.

A point estimate reported alone tells a reader nothing about how far it might have landed from the truth. Ours landed 0.2227 away.

What the Interval Is Made Of

Three numbers and nothing else: the estimate 1.5227, its standard error 0.1897, and a critical value read off the t distribution on 118 degrees of freedom. Two of the three were on the board at the end of Lecture 2.3.

Deriving the Interval From the t Distribution

The distribution of the standardised slope is a statement about a quantity we cannot compute. Three lines turn it into two numbers we can:

$$\begin{aligned} 0.95 &= \Pr\left(-c < \frac{\hat{\beta}_1 - \beta_1}{\text{se}(\hat{\beta}_1)} < c\right) \\ &= \Pr(-c \text{ se}(\hat{\beta}_1) < \hat{\beta}_1 - \beta_1 < c \text{ se}(\hat{\beta}_1)) \\ &= \Pr(\hat{\beta}_1 - c \text{ se}(\hat{\beta}_1) < \beta_1 < \hat{\beta}_1 + c \text{ se}(\hat{\beta}_1)) \end{aligned} \tag{4}$$

The first line is the definition of the critical value. The second multiplies through by the standard error, which is positive, so the inequalities keep their direction. The third subtracts $\hat{\beta}_1$ and multiplies by -1 , which reverses them and swaps the two ends (Wooldridge, 2020).

Only the Endpoints Differ From Sample to Sample

β_1 is one fixed number in all three lines and is never the thing that varies. What differs from one sample to the next is $\hat{\beta}_1$, and therefore both endpoints. The last line is the interval.

The Interval on Our 120 People

The three numbers, multiplied out:

$$1.5227 \pm 1.980 \times 0.1897 = 1.5227 \pm 0.3756 = [1.147, 1.898] \quad (5)$$

- 0.3756: the half-width, which is the standard error scaled by the critical value

The critical value is 1.980 rather than Lecture 1.3's 1.96 because 118 degrees of freedom is a t and not a normal. Above 120 degrees of freedom Wooldridge licenses 1.96, and above 50 he licenses a plain two standard errors (Wooldridge, 2020).

The Width Comes From the Standard Error, Not the Estimate

1.5227 only says where the interval sits. Its width is 0.1897 scaled, and so it carries the three ingredients of Lecture 2.3: the error variance, the spread of schooling, and the number of people.

The Interval Covers the True Slope

The population slope is 1.30, our estimate is 1.5227, and the interval runs from 1.147 to 1.898. The true value sits inside it, and not at its edge.

Two things are true at once, and this is the only place in the module where both can be checked against a known truth:

- **The point estimate missed.** $1.5227 - 1.30 = 0.2227$, and no arithmetic error produced that.
- **The interval caught it.** The machinery of 2.3 and today worked on this sample exactly as advertised.

In any real application the first is unknowable and the second unverifiable. Here we wrote the population, so we can look.

This Is What the Machinery Was Built to Do

An unbiased rule that misses by 0.2227 is not a failure, provided the number printed beside it is honest about how far it might have missed. Reporting 1.5227 alone conceals that. Reporting 1.147 to 1.898 does not.

Reading the Interval Out Loud

The wording is examinable, and the indefensible version sounds better than the defensible one. Lecture 1.3 drew this distinction for a mean; for a slope the sentence has to carry units as well.

The sentence to avoid is “there is a 95 per cent chance the true return lies between 1.147 and 1.898”. It makes β_1 the thing that varies, and β_1 is one fixed number in the population of 200,000.

The Sentence to Use

Our estimate of the return to a year of schooling is **1.5227 euro an hour**, with a 95 per cent confidence interval from **1.147 to 1.898**. Comparing two people in this sample whose schooling differs by one year, that is the earnings gap the line reports.

What the Ninety-Five Per Cent Counts

It counts intervals, not truths. Lecture 1.3 established this for a mean and nothing about a slope is different (Wooldridge, 2020):

- **Before the sample is drawn:** the rule produces an interval covering β_1 in 95 per cent of samples.
- **After the endpoints are numbers:** 1.30 is either inside 1.147 to 1.898 or it is not.

In this module we can see which, and on the previous frame we did. In any application you run, you cannot: Wooldridge's own wording is that we hope our sample is one of the 95 per cent where the interval contains the truth, and have no guarantee.

The Coverage Belongs to the Procedure, and the Procedure Includes the Sampling

The 95 per cent is a property of the rule under assumptions one to five. If the 120 chose themselves, or if earnings spread out more at higher schooling, the rule covers less often, and the printed interval will not say so (Wooldridge, 2020).

The Interval and the Test Are One Statement

A value is rejected at five per cent exactly when it falls outside the 95 per cent interval. The two are one inequality written twice (Wooldridge, 2020):

- **Zero is outside 1.147 to 1.898**, so a null of zero is rejected, which is what $t = 8.03$ said.
- **1.30 is inside it**, so a null of 1.30 is not rejected, which answers the question asked in section 1.

The interval settles both at once, and it settles the same question for every other candidate value as well. That is the argument for printing it rather than printing a verdict.

An Interval Is Every Test You Could Have Run

Each value inside the interval is a null this sample does not reject, and each value outside it is one it does. A single test reports one of those verdicts and throws the rest away.

The Interval Against the Sampling Distribution

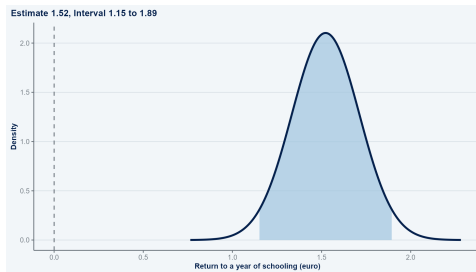


Figure 1: The distribution implied by 1.5227 and 0.1897, with the 95 per cent region shaded.

The Shaded Region Excludes Zero and Contains 1.30

The curve is centred on our estimate, because an interval is built around what we have rather than around what is true. Zero sits far outside the shaded region. The true 1.30 sits inside it, left of centre by 0.2227.

A Wide Interval or a Narrow One

Two research teams report the return to a year of schooling. One estimates 1.5, with an interval from -0.1 to 3.1 . The other estimates 0.4 , with an interval from 0.3 to 0.5 .

Question

Which of the two would you rather have been given, and does your answer agree with which of the two is statistically significant?

SECTION 3

Reading a Regression Table

1. Testing a Coefficient
2. Confidence Intervals for an Effect
3. Reading a Regression Table

The Same Estimate at Four Different Sample Sizes

This is a thought experiment, not four samples. The estimate is frozen at the 1.5227 our 120 people happened to give, and only the number of people differs. A real sample of 480 would return its own estimate, usually one nearer 1.30.

People	Slope	Standard error	t	95 per cent interval
30	1.5227	0.3843	3.96	[0.735, 2.310]
120	1.5227	0.1897	8.03	[1.147, 1.898]
480	1.5227	0.0946	16.10	[1.337, 1.709]
1920	1.5227	0.0472	32.23	[1.430, 1.615]

Four times the people halves the standard error and doubles the t , which is Lecture 2.3's $\sqrt{n-1}$: $1.5227/0.1897 = 8.03$ and $1.5227/0.0472 = 32.23$, with the same estimate on top of both.

The Last Two Intervals Do Not Cover 1.30

They exclude the truth because the estimate was frozen at a number that had already missed it by 0.2227. A narrow interval around an estimate that missed is confidently wrong: precision is not accuracy, and a t of 32.23 is no evidence that 1.5227 sits near 1.30.

One Return to Schooling, Four Sample Sizes

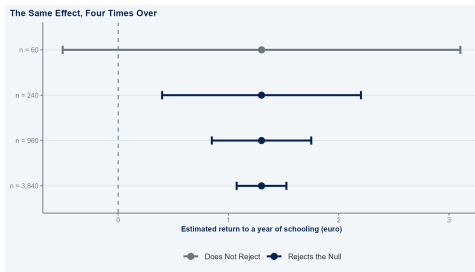


Figure 2: The same estimated return to schooling, reported at four different sample sizes.

All Four Exclude Zero and the Two Narrowest Exclude 1.30

Every interval is drawn around the same frozen estimate, so the four share a centre and differ only in width. Each one rejects a null of zero. The two narrowest also reject the true value, and no real study could know that.

Why Significance Is About Precision, Not Effect Size

Significance is a statement about how precisely something was measured. It is not a statement about how large it is, and the two are confused in print constantly (Wooldridge, 2020).

The mechanism is visible in the ratio. A t has the estimate on top and a precision underneath, so a large t comes from a large estimate or from a small standard error, and a star alone cannot tell you which:

- **A very large sample** makes a negligible return significant, because the standard error is tiny.
- **A small sample** misses a return that matters, because a wide standard error raises the bar.

Wooldridge's 401(k) example has 1,534 firms, every coefficient significant at small levels, and one of them practically trivial (Wooldridge, 2020).

Practical Significance Is a Separate Judgement

Whether 1.5227 euro an hour per year of schooling is worth acting on is a question about Irish wages, 21

What to Read First in a Regression Table

Our own regression, laid out the way a published table lays one out:

	Hourly earnings
Years of schooling, centred at 13	1.5227
Standard error	(0.1897)
95 per cent interval	[1.147, 1.898]
t against a null of zero	8.03
People	120

Only the first two lines are printed by most tables. The interval is the one line a reader can do most with, and it is the one most often left out.

The Order to Read Them In

The coefficient in units first, then the standard error, then the sample size and what it is a sample of, then what else is in the regression, which is Lecture 3.1's subject. The stars last, and only if the first four are there.

What a Table of Stars Leaves Out

A table printing coefficients and stars, with no standard errors, has removed the number a reader most needs. A star says only that $|t|$ passed a threshold, so our 8.03 and a t just past 1.980 carry the same mark. Three consequences:

- **No interval can be rebuilt**, so no reader can ask whether 1.30, or any other value, is consistent with the data.
- **A precise small estimate reads exactly like an imprecise large one**, which is the previous frame in one line.
- **The result cannot be combined with anybody else's**, because combining estimates needs their standard errors.

Report the Interval in Your Own Tables

The interval carries the estimate and its precision together, and the test is inside it. When the table is yours, print 1.147 to 1.898 beside 1.5227 and a reader needs nothing further from you.

What Would Have To Be True?

This time the question is asked of the interval we just built, 1.147 to 1.898, and of the sentence we agreed to report it in.

Question

What would have to be true for this number to mean what we just said it means? And what is the plausible story where it isn't?

Application

The app reports one estimate four ways, bare, with a standard error, with an interval and with a star, and it raises the sample size while holding the estimate at 1.5227.

Question

If the sample size goes to 480 and the estimate stays at 1.5227, which of the four printed forms differ, and which of the four does a reader need?

The lab for this lecture is `confint()` beside `summary()`, and building a regression table that prints standard errors rather than stars.

Summary: Testing, Intervals and Tables

1 Testing a Coefficient.

$t = 1.5227/0.1897 = 8.03$ on 118 degrees of freedom, against a five per cent critical value of 1.980.

2 Confidence Intervals for an Effect.

$1.5227 \pm 1.980 \times 0.1897$ runs from 1.147 to 1.898, and it covers the true 1.30.

3 Reading a Regression Table.

The coefficient in units first, the standard error second, and the stars last if at all.

Next Lecture

Lecture 3.1 puts a second variable into the regression, and it is the first lecture that returns a different number in place of 1.5227:

- **Partialling Out:** what adding a second variable does to the slope you already had.
- **Reading a Partial Coefficient:** what holding something fixed covers, and three variables where it goes wrong.
- **Omitted Variable Bias:** turning the thing you could not measure into a number.

Everything reported today rests on assumptions one to five. Lecture 3.1 is the first lecture in which one of them is false on purpose.

Before Lecture 3.1

1 **Play:**

The app. Hold the estimate at 1.5227, raise the sample size, and watch the interval narrow around it.

2 **Apply:**

Problem set 2, covering 2.3 and 2.4. It rebuilds today's interval by hand from a standard error you compute yourself.

3 **Review:**

Wooldridge (2020), sections 4-1 to 4-3, on the t test and confidence intervals.

Thank you

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BACKUP

Appendix

Derivation: Why the Distribution Is t and Not Normal

The body frames asserted the t. With σ known, the standardised slope would be standard normal, because a weighted sum of normal errors is normal:

$$\frac{\hat{\beta}_1 - \beta_1}{\text{sd}(\hat{\beta}_1)} \sim \text{Normal}(0, 1) \qquad \frac{\hat{\beta}_1 - \beta_1}{\text{se}(\hat{\beta}_1)} \sim t_{118} \qquad (6)$$

One Estimated Number Separates the Two

The only difference is that σ on the left has been replaced by $\hat{\sigma}$ on the right, and $\hat{\sigma}$ was itself computed from the 120 residuals in Lecture 2.3. That second source of variation thickens the tails, which is why 1.980 is larger than 1.96 (Wooldridge, 2020).

Derivation: The Interval as the Set of Nulls That Survive

Run section 1's test at every possible null value a in turn and keep the values the sample does not reject:

$$\left| \frac{\hat{\beta}_1 - a}{\text{se}(\hat{\beta}_1)} \right| \leq c \quad \Longleftrightarrow \quad \hat{\beta}_1 - c \text{se}(\hat{\beta}_1) \leq a \leq \hat{\beta}_1 + c \text{se}(\hat{\beta}_1) \quad (7)$$

The Same Two Endpoints by a Second Route

The values that survive are exactly 1.147 to 1.898. This is why a null is rejected at five per cent precisely when it falls outside the 95 per cent interval, and why the two frames in section 2 were making one claim (Wooldridge, 2020).

Notes and Tools

- **The App:** the reporting app, with the sample size raised while the estimate is held fixed.
- **The Lecture in One Line:** `confint(lm(y ~ I(x - 13)), level = 0.95)`.
- **This Lecture's Lab:** `summary()` and `confint()`, and one table package.

References

Angrist, J. D., & Pischke, J.-S. (2015). *Mastering 'metrics: The path from cause to effect*. Princeton University Press.

Wooldridge, J. M. (2020). *Introductory econometrics: A modern approach* (7th ed.). Cengage Learning.